

BIOS 755: Model Diagnostics

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First, let's review model diagnostics for standard regression models

Two model diagnostic topics that are of interest:

1. Examine whether the assumptions of linear regression models satisfied
 - ▶ Linear, Independent, Normality, and Equal variance (LINE)
 - ▶ Most of the assumptions of linear regression can be checked by plotting the estimated residuals

$$e_i = Y_i - \hat{Y}_i$$

2. Identify points that potentially influence the fitted regression line (outliers and influential points)

Line

- ▶ Assumption: there is a linear relationship between Y and the X 's. I.e.,
 - ▶ The conditional mean of Y is correctly specified as linear in the parameters.
 - ▶ The expected value of the residual is zero for all combinations of X 's.
 - ▶ $E(e_i) = 0$.
- ▶ How to check
 - ▶ Plot Y_i vs. X_i .
 - ▶ Plot e_i vs. $\hat{Y}_i$

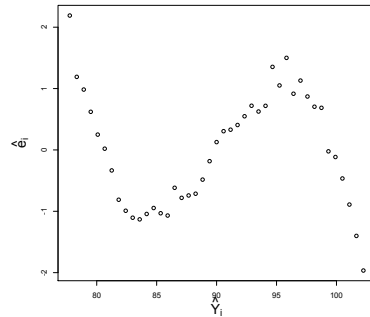
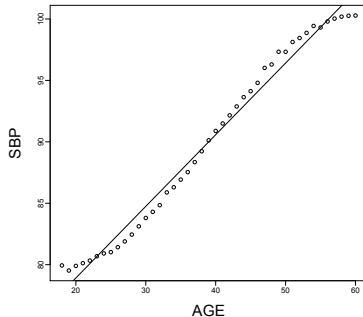
Line

Example Violation:

- ▶ Systematic curvature in residual plots indicates model misspecification of the mean structure.

Corrective actions:

- ▶ Fit a different regression model.
- ▶ Add quadratic of cubic components.
- ▶ Transformation of Y and/or X



Normality

- ▶ Assumption: The residuals are normally distributed.
- ▶ In large samples, normality of residuals is less critical for inference due to asymptotic theory.
- ▶ How to check
 - ▶ Histogram of the e_i 's.
 - ▶ Normal probability plot of e_i .
 - ▶ Tests of normality on e_i .
 - ▶ Outlier tests.

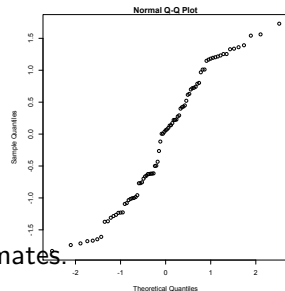
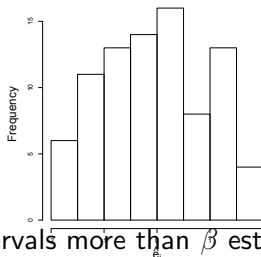
Normality

Corrective actions:

- ▶ Examine outliers to determine if they are contaminated
- ▶ Transformation of Y and/or X

Notes:

- ▶ Skewness $\rightarrow$ affects prediction intervals more than β estimates.
- ▶ Heavy tails $\rightarrow$ inflate Type I error in small samples.

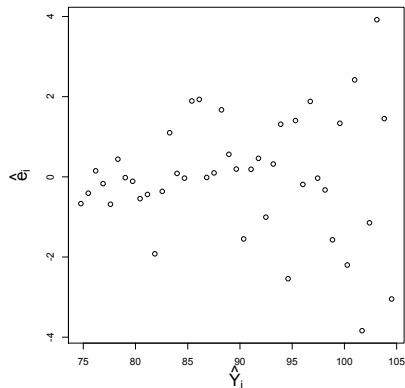


Equal Variance

- ▶ **Assumption:** the variance of the residuals is equal for all X 's.
- ▶ How to check
 - ▶ Plot Y_i vs. X_i .
 - ▶ Plot e_i vs. $\hat{Y}_i$

Example Violation $\Rightarrow$

- ▶ **Corrective action:**
 - ▶ Transformation of Y .
 - ▶ **Patterned covariance model**



If an assumption is violated, what is the effect?

- ▶ Line:
 - ▶ Predictive estimates ($\hat{Y}_i$) are biased estimates of $E(Y_i)$. Estimates of β are biased.
 - ▶ **Predictive intervals** of Y_i are not valid.
 - ▶ *Confidence intervals and hypothesis tests* of β , $\hat{Y}_i$ are not valid.
- ▶ Independent:
 - ▶ **Predictive intervals** of Y_i are not valid.
 - ▶ *Confidence intervals and hypothesis tests* of β , $\hat{Y}_i$ are not valid.
- ▶ Normality:
 - ▶ **Predictive intervals** of Y_i are not valid.
 - ▶ *Confidence intervals and hypothesis tests* of β , $\hat{Y}_i$ are not valid for small samples (i.e., $n < 30$).
- ▶ Equal variance:
 - ▶ **Predictive intervals** of Y_i are not valid.
 - ▶ *Confidence intervals and hypothesis tests* of β , $\hat{Y}_i$ are not valid.

Are these a problem? Depends on your question.

Linear Mixed Model

- ▶ The linear mixed model can be expressed as

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i + \mathbf{e}_i$$

where

- ▶ $\mathbf{X}_i$ – $n_i \times p$ matrix of covariates
- ▶ $\mathbf{Z}_i$ – $n_i \times q$ matrix of covariates (usually the columns of $\mathbf{Z}_i$ are a subset of the columns of $\mathbf{X}_i$ and $q < p$).
- ▶ $\boldsymbol{\beta}$ – $p \times 1$ vector of fixed effects.
- ▶ $\mathbf{b}_i$ – $q \times 1$ vector of random effects, $\mathbf{b}_i \sim N(0, \mathbf{G})$,
- ▶ $\mathbf{e}_i$ – $n_i \times 1$ vector of errors and $\mathbf{e}_i \sim N(0, \mathbf{R}_i)$.

Types of residuals

Four types of residuals

► **Unconditional residuals**

$$r_i^u = Y_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}$$

These are not as good for model diagnostics.

► **Conditional residuals**

$$r_i^c = Y_i - (\mathbf{X}_i \hat{\boldsymbol{\beta}} + \mathbf{Z}_i \hat{\mathbf{b}}_i)$$

where $\hat{\mathbf{b}}_i$ are the EBLUP random effects.

► **Scaled residuals**

$$r_i^s = \hat{\mathbf{L}}_i^{-1} r_i^u$$

where $\hat{\boldsymbol{\Sigma}}_i = \hat{\mathbf{L}}_i \hat{\mathbf{L}}_i'$ and $\hat{\boldsymbol{\Sigma}}_i = \mathbf{Z}_i \hat{\mathbf{G}} \mathbf{Z}_i' + \mathbf{R}_i$.

Types of residuals

▶ Studentized conditional residuals

- ▶ Each conditional residual is divided by its estimated standard deviation

$$r_i^{st} = \frac{r_i^c}{\sqrt{\text{Var}(r_i^c)}}$$

- ▶ By scaling the residuals, we can remove the effect of unequal variance (i.e., by time point) in our model.
 - ▶ These are appropriate to use for linear mixed models.
- ▶ Use studentized conditional residuals for LMMs.
- ▶ For patterned covariance models, use scaled residuals.

Checking Assumptions in Longitudinal Models: What Changes?

- ▶ In OLS regression we check LINE (Linearity, Independence, Normality, Equal variance).
- ▶ In longitudinal models, **independence is replaced by covariance specification**.
- ▶ Practical checklist:
 1. **Mean structure:** is the time trend and covariate functional form reasonable?
 2. **Covariance structure:** does the chosen pattern match the data (e.g., does correlation decay with lag)?
 3. **Residual behavior:** are residuals roughly centered at 0; any systematic patterns vs. fitted values or time?
 4. **Normality (if using LMM):** are (appropriate) residuals approximately normal? (More important in small samples.)
 5. **Variance by time:** does variability change across visits? If so, consider heterogeneous covariance structures.

Longitudinal Diagnostics: What to Plot

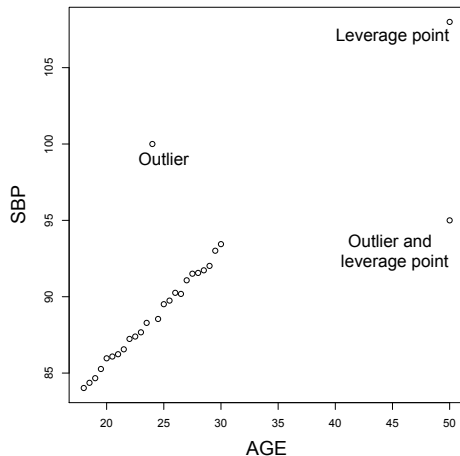
- ▶ **Spaghetti plots:** look for nonlinearity, heterogeneity, outliers, and missingness patterns.
- ▶ **Mean profiles over time (by group):** check whether linear time is reasonable or if curvature suggests quadratic/spline/profile time.
- ▶ **Residual plots:**
 - ▶ residual vs. fitted (mean misspecification / nonlinearity),
 - ▶ residual vs. time (time-specific patterns),
 - ▶ residual variance vs. time (heteroscedasticity by visit).

Introduction

We will discuss the detection of **outliers** and **leverage points**.

What's the difference?

- ▶ **Leverage point:** unusual predictor combination, i.e. $(X_{i1}, \dots, X_{ip})$.
- ▶ **Outlier:** Unusual Y_i values.
- ▶ High leverage does not imply large influence unless the residual is also large.



How to identify leverage?

How do we detect X_i values with large leverage?

- ▶ Calculation of leverage of X_i is done by h_{ii} is the diagonal elements of the so-called hat matrix H ($H = X'(X'X)^{-1}X$)
- ▶ The greater the h_{ii} , the greater the leverage
- ▶ The both high and low $\mathbf{X}_i$ compared to the sample mean $\bar{\mathbf{X}}$ gives large leverage.
- ▶ Leverage tells us how strange the predictor variables are.
- ▶ Leverage (and in general influence) can be estimated on a subject or observation level.
 - ▶ **Observation-level:** one time point for one subject.
 - ▶ **Subject-level (cluster-level):** an entire trajectory.

How to identify influence?

- ▶ The basic procedure for quantifying influence is simple:
 1. Fit the model to the data and obtain estimates of all parameters.
 2. Remove one or more data points from the analysis and compute updated estimates of model parameters.
 3. Based on full- and reduced-data estimates, contrast quantities of interest to determine how the absence of the observations changes the analysis.
- ▶ We will look at the influence an observation has on the **overall** model fit, as well as on **mean and variance** parameter estimates.

Overall influence

- ▶ The overall influence of an individual on the model fit can be measured by the change in the likelihood when the individual is not included.
- ▶ This is measured by the likelihood and restricted likelihood distances

$$\begin{aligned}LD_i &= 2\{l(\hat{\theta}) - l(\hat{\theta}_{-i})\} \\ RLD_i &= 2\{l_R(\hat{\theta}) - l_R(\hat{\theta}_{-i})\}\end{aligned}$$

- ▶ The likelihood distance is a global summary measure, expressing the joint influence of the individual on the parameter estimates of θ

Change in Parameter Estimates

- ▶ Change in the parameter estimates can be estimated in a similar manner.
- ▶ **Cook's D** and the multivariate **DFFITS** statistic are two measures of how much the parameter estimates change when an individual is removed from the analysis.
- ▶ While we won't go over the specific formulas, we're concerned about large values, indicating that the change in the parameter estimate is large relative to the estimate's variability.

Change in Parameter Estimates Variance

- ▶ Another way to measure the influence of an individual is to look at how the variance of parameter estimates changes.
- ▶ Data points that have a small Cook's D, for example, can still greatly affect hypothesis tests and confidence intervals.
- ▶ **COVTRACE** and **COVRATIO** are two quantities in SAS that measure the impact on the covariance estimates.
- ▶ The benchmarks of “no influence” are:
 - ▶ Zero for **COVTRACE**
 - ▶ One for **COVRATIO**

Influence Measures (summary)

- ▶ **Overall influence (likelihood-based):**
 - ▶ Likelihood distance (LD), restricted likelihood distance (RLD): change in (restricted) likelihood when a subject is removed.
- ▶ **Influence on parameter estimates:**
 - ▶ Cook's D, DFFITS: how much fixed-effect estimates change when removing a unit.
- ▶ **Influence on precision:**
 - ▶ COVTRACE and COVRATIO: impact on the covariance matrix of parameter estimates.
- ▶ Practical use: treat these as **screening tools**.

Practical threshold guideline

What is large?

- ▶ Cook's $D > 4/n$ (rule of thumb for regression)
- ▶ Large relative to others
- ▶ Visual screening preferred

When You Detect an Outlier: First Steps

- ▶ An **outlier** is an observation (or subject) that does not follow the general pattern of the data.

Step 1: Verify data integrity

- ▶ Check units, coding errors, impossible values.
- ▶ Confirm time ordering and visit labels.
- ▶ Look for duplicate or miscoded records.

Step 2: Classify the type of outlier

- ▶ **Outcome outlier:** unusual Y given covariates and time.
- ▶ **Covariate outlier:** unusual X (may create leverage).
- ▶ **Subject-level outlier:** an entire trajectory looks atypical.

Key question: Is this a data problem, or a real but extreme observation?

From Outlier to Influence: Does It Change Conclusions?

- ▶ An outlier becomes **influential** if removing it materially changes:
 - ▶ Fixed effect estimates,
 - ▶ Standard errors or p-values,
 - ▶ Covariance parameter estimates.

Step 3: Assess influence

- ▶ Compare model results with and without the point (or subject).
- ▶ Determine whether influence is observation-level or subject-level.
- ▶ Identify whether leverage, large residuals, or sparse follow-up drive the effect.

Step 4: Decide on action

- ▶ Remove only if there is a clear data-quality issue.
- ▶ Otherwise, report sensitivity analyses.
- ▶ Interpret results cautiously and transparently.

Take-home: Not all outliers are influential — and not all influential points should be removed.

Removing Data Points: Pros and Cons

Potential benefits

- ▶ Removes obvious data errors/contamination.
- ▶ Can stabilize estimation when a small number of points dominate.
- ▶ Improves model fit when violations are driven by a few erroneous records.

Potential harms

- ▶ Can introduce bias if removals are related to exposure/outcome.
- ▶ Can reduce generalizability by excluding true extremes.
- ▶ Encourages “p-hacking” if done without a pre-specified rule.
- ▶ May hide model misspecification (the model is wrong, not the data).

Best practice: remove points only for clear data quality reasons; otherwise, keep them and use sensitivity analyses and transparent reporting.

Summary: Outliers vs. Leverage vs. Influence

- ▶ **Outlier:** unusual Y value given the model (large residual).
- ▶ **Leverage point:** unusual X profile (unusual covariate combination).
- ▶ **Influential point:** materially changes estimates/inference when removed.
- ▶ Relationships:
 - ▶ A point can be an outlier but not influential (large residual, low leverage).
 - ▶ A point can have high leverage but not be influential (unusual X , small residual).
 - ▶ The most influential points often combine **high leverage + large residual**.
- ▶ In longitudinal data, influence is often **subject-level** rather than single-observation-level.

Regression Diagnostics: Big Picture Workflow

1. **Start with plots:** outcome vs. predictors; spaghetti/mean profiles (longitudinal).
2. **Fit a reasonable mean model:** choose time functional form and key covariates.
3. **Check residual behavior:**
 - ▶ residual vs. fitted (mean misspecification),
 - ▶ residual vs. time (time-related patterns),
 - ▶ distribution of residuals (normality, tail behavior),
 - ▶ variance patterns (heteroscedasticity).
4. **Address correlation (longitudinal):** pick/validate covariance structure.
5. **Screen for outliers/influence:** identify points/subjects that drive results.
6. **Sensitivity analyses + reporting:** refit under reasonable alternatives and document decisions.

Diagnostics: What To Do When Something Looks Wrong

- ▶ **Nonlinearity:** consider transformations, polynomial terms, splines, or profile (categorical) time.
- ▶ **Heteroscedasticity:** consider transformations; in longitudinal models consider heterogeneous covariance (e.g., CSH).
- ▶ **Non-normal residuals:** check outliers/contamination; consider transformations or robust approaches (especially in small samples).
- ▶ **Correlation misspecification:** try alternative covariance patterns; compare AIC/BIC with mean model fixed; inspect implied correlations.
- ▶ **Outliers/influential points:** validate data, understand mechanism, and run sensitivity analyses rather than deleting reflexively.

Take-home: diagnostics are about protecting your scientific conclusions, not achieving “perfect” plots.